(1)

Aim:- To build a system to break the RSA encryption Ciphers.

Sample Size Considered: 3000

X Matrix (It is considered as Feature Matrix):

* The features to break the RSA encryption is considered as:-
  - → Cipher text length
  - → frequancy of occurence of Identifiable patterns
  - → size of Public key
  - → Timing Information
  - → public Exponent (e)

In X-matrix Each row is considered to sample and coloumn is considered to feature of the data.

Dimensions :-

so for the considered sample the dimensions are $3000 \times n$ where n = no.of.features

As here we considered 5 features in this problem. the dimension of X-Matrix would be $m \times n$ i.e, 3000×5

Matrix representation:-

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{3000,1} & x_{3000,2} & \cdots & x_{3000,n} \end{bmatrix}$$

here x_{ij} represents jth feature of ith sample.

y - vector :- [considered as Output Vector] :-

y-vector represents target variable or the class labels which the model is trying to predict.

So, here we clarify the Output vector in binary format which helps to identify whether the sample can succesfully break the RSA encryption ciphers or not.

∴ Classes :- (Output values :)

0 : Cannot break RSA encryption ciphers

1 : Can break RSA encryption ciphers

∴ Dimensions :- y vector will have 3000 X 1, which represent binary classification for every sample.

vector representation :-

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{3000} \end{bmatrix}$$

here y_i can be either 0 or 1.

(2) Aim! - To build a system the detects debris in outer space.

Considered Sample size! - 5000

X - Matrix! - (Feature Matrix)

Each row represent different sample and
Each coloumn represent diffent feature.

Considered Features!-

* object size
* Reflectivity
* Trajectory data
* velocity
* Distance from sensor
* shape Analysis.

∴ Dimensions! - X matrix will have : $5000 \times n$,

where $5000 \rightarrow$ sample size
$n \rightarrow$ no. of features

Considere Example features $n=6$:

∴ Dimension of X matrix would be 5000×6

∴ Matrix Representaion!-

$$X = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{5000,1} & x_{5000,2} & \cdots & x_{5000,n} \end{pmatrix}$$

X_{ij} represents the j-th feature of ith sample.

* y-vector:- (output vector)

represent target variable

classes:- (out put values which are considered with binary classification)

0: Not a debris

1: Debris.

vector representation:-

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{5000} \end{bmatrix}$$

y_i can be either 0 or 1. which indicates wher i^{th} sample is debris (or) not.

(3) Aim:- To build a system to detect faulty parts of Conveyor belt.

Considered sample size:- 5000

X-Matrix:- (Feature Matrix)

→ considered features:-

* Weight
* Dimensions
* shape deviation
* Assembly fit
* vibration signature.

n = no. of features
m = sample size (5000)

∴ The dimension matrix would be 5000 x n

here according to considered features the dimension matrix would be 5000 X 5 (here n=5)

$\downarrow$ no. of features considered.

General Matrix Representation:-

$$X = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{5000,1} & x_{5000,2} & \cdots & x_{5000,n} \end{pmatrix}$$

y-vector:- (output vector)

classes:- 0 - Non faulty part
1 - faulty part

Dimensions:-

y vector will have dimensions of 5000×1

where 5000 represent samples

& 1 represent binary classification

vector representation:-

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{5000} \end{bmatrix}$$

y_i represent the classification of i^{th} sample

is came eithe 0 (or) 1

(4) Aim:- To build a system to generate images based on descriptive text.

Sample size considered : 10,000

X-Matrix:- (feature Matrix)

Considered features:-

* Word count
* Average word length
* Presence of objects
* sentence sentiment
* key themes (or) topics in sentence
* No.of Adjectives

Dimensions:-

Sample size 10,000

n = no.of features

Considere features for the above proble n=6

$\therefore$ 10,000 X n i.e 10,000 X 6

Matrix representation:-

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ x_{10000,1} & x_{10000,2} & \cdots & x_{10000,n} \end{bmatrix}$$

where X_{ij} represent jth feature of ith descriptive sentence.

y - vector :- (output vector.)

$$\therefore y = \begin{bmatrix} y_1 \cdots y_{1m} \\ y_2 \cdots y_{2m} \\ \vdots \\ y_{10{,}000} \, y_{10{,}000,m} \end{bmatrix}$$

here y_{ij} ~~canbe~~ represent jth pixel value of ith image

classes ! -

The output here is continuous rather than categorical, with each value corresponding to pixel intensities of image